

Capturing traffic over Wi-Fi Voice Commands Privacy Analysis of Augmented Reality Glasses

1 Introduction

This experiment captures the Classic Bluetooth traffic whilst doing various voice commands. The capture focused on interactions between the glasses and phone, which was connected to the internet. The goal was to analyze the different voice commands done on the glasses in regards to generating Classic Bluetooth packets. By examining this traffic, the aim is to understand the communication patterns.

2 Metadata

Time of experiment: Nov 7th 13:55

Duration of experiment: Roughly 1 hour

Addresses:

3 Voice Commands used

- Power on glasses
- Re-pair glasses to phone
- "Hey Meta, what's the weather in [place] today?"
- "Hey Meta, what time is it?"
- "Hey Meta, what date is today?"
- "Hey Meta, what's the battery percentage?"

- "Hey Meta, play music."
- "Hey Meta, increase the volume."
- "Hey Meta, lower the volume."
- "Hey Meta, pause music."

- "Hey Meta, take a photo."
- "Hey Meta, take a video."
- "Hey Meta, stop recording."

- "Hey Meta, send a message to [name]."
- "Hey Meta, send a voice message to [name]."
- "Hey Meta, read my messages."

- "Hey Meta, set a timer for 1 minute."
- "Hey Meta, how much time is left in my timer."
- "Hey Meta, cancel my timer for 1 minute."

4 Time Stamps

5 Note

Wanted to wait till media finished upload so the experiment extended from 25 minutes to an hour. I tried to take the glasses out of case to wear for import and stopped media import by taking another photo. I got an 'error occurred during import'. It had to reconnect and restart import. However, the import was still not done by the time I stopped capture. Something interesting is that when I open the glasses, I see a reaction on the terminal. However, this was not visible on Wireshark.

The command for the weather was unsuccessful.

After looking on Wireshark, only 2 packets were visible. This is interesting as during the capture, there were more than 2 packets and an experiment of one hour where half of it is with continuous interaction should have more than 2 packets.